

Code

05 July 2023 15:11

1. Cd WPS_Path
2. Change namelist.wps
3. Check area:

```
ncl util/plotgrids_new.ncl
```

4. Deal with Geographical static data:
./geogrid.exe

```
Processing OA3SS
Processing OA4SS
Processing OL1SS
Processing OL2SS
Processing OL3SS
Processing OL4SS
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
! Successful completion of geogrid. !
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

```
ncview geo_em.d01.nc #(check output)
```

5. Deal with meteorological data: (use v83 file* <https://rda.ucar.edu/datasets/ds083.2/dataaccess/>)

```
cd ../../data
python download_ds094.1.py #(download data)
python download_ds083.2.py #(download data)
```

```
./NCEP_rename_1.sh #(rename file)
./NCEP_rename_1.sh #(rename file)
```

```
cd ungrib
./g2print.exe ../../../../data/NCEP/fnl.2020-01-01_00.grib2 #(check data format)
```

NCEP CDAS CFSV2											
rec num	Prod Disc	Cat num	Param num	Lvl code	Lvl one	Lvl two	Prod Templ	Name	Time	Fcst hour	
1	0	0	0	1	0	0	0	TMP	2019-12-31_18:00:00	06	
2	2	0	192	106	0	10	0	SOILW	2019-12-31_18:00:00	06	
3	2	0	192	106	10	40	0	SOILW	2019-12-31_18:00:00	06	
4	0	0	0	106	0	10	0	TMP	2019-12-31_18:00:00	06	
5	0	0	0	106	10	40	0	TMP	2019-12-31_18:00:00	06	
6	0	1	13	1	0	0	0	WEASD	2019-12-31_18:00:00	06	
7	2	0	0	1	0	0	0	LAND	2019-12-31_18:00:00	06	
8	10	2	0	1	0	0	0	ICEC	2019-12-31_18:00:00	06	
10	0	2	2	103	10	0	0	UGRD	2019-12-31_18:00:00	06	
10	0	2	3	103	10	0	0	VGRD	2019-12-31_18:00:00	06	
11	0	0	0	103	2	0	0	TMP	2019-12-31_18:00:00	06	
12	0	1	0	103	2	0	0	SPFH	2019-12-31_18:00:00	06	
13	0	3	0	1	0	0	0	PRES	2019-12-31_18:00:00	06	
14	0	3	196	1	0	0	0	HPBL	2019-12-31_18:00:00	06	
15	2	0	192	106	40	100	0	SOILW	2019-12-31_18:00:00	06	
16	2	0	192	106	100	200	0	SOILW	2019-12-31_18:00:00	06	
17	0	0	0	106	40	100	0	TMP	2019-12-31_18:00:00	06	
18	0	0	0	106	100	200	0	TMP	2019-12-31_18:00:00	06	
19	0	3	5	1	0	0	0	HGT	2019-12-31_18:00:00	06	

Successful completion of g2print

```
./link_grib.csh ../../../../data/NCEP/ #(link file) (output: GRIBFILE*)
./ungrib.exe #(convert data format) (output: fnl*)
```

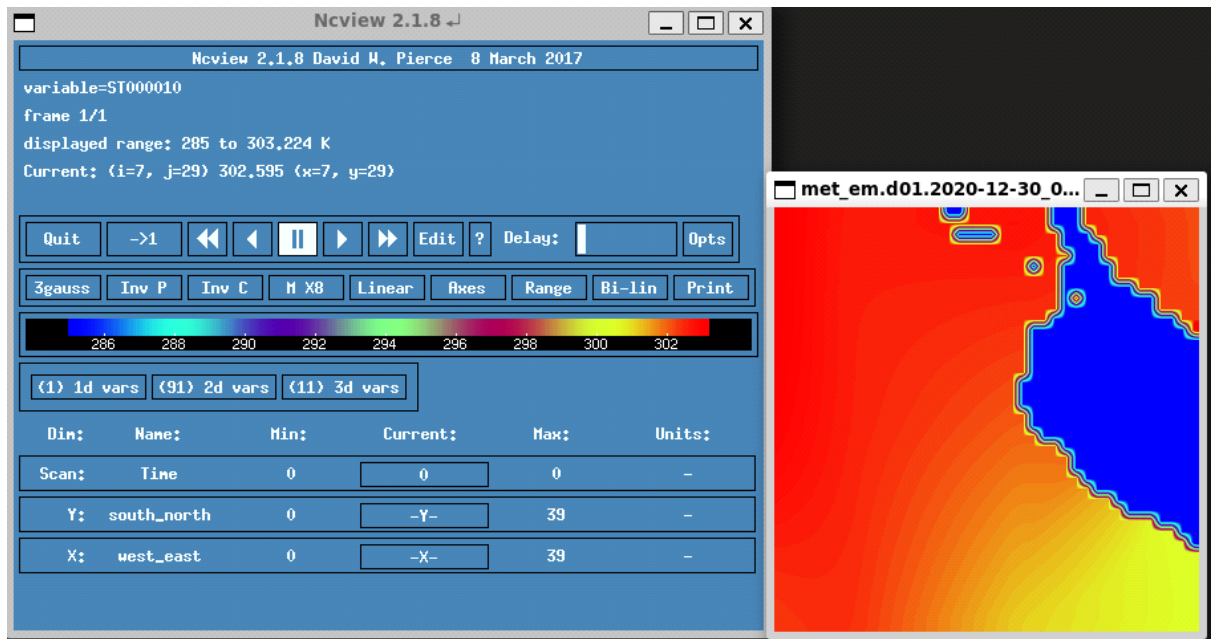
```
Deleting file: ./PFILE:2020-12-31_12
Deleting file: ./PFILE:2020-12-31_18
Deleting file: ./PFILE:2021-01-01_00

*****
Done deleting temporary files.
*****

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
! Successful completion of ungrib. !
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

6. Generate met* file

```
./metgrid.exe #(output: met_em.d01*)
ncview met_em.d01.2020-12-30_00\00\00.nc
```



```
ncdump -h met_em.d01.2020-01-01_00\:\00\:\00.nc >& log.ncdump #(Check the number of levels data has)
```

7. Generate initial wrfinput_* file (file namelist_init.input)

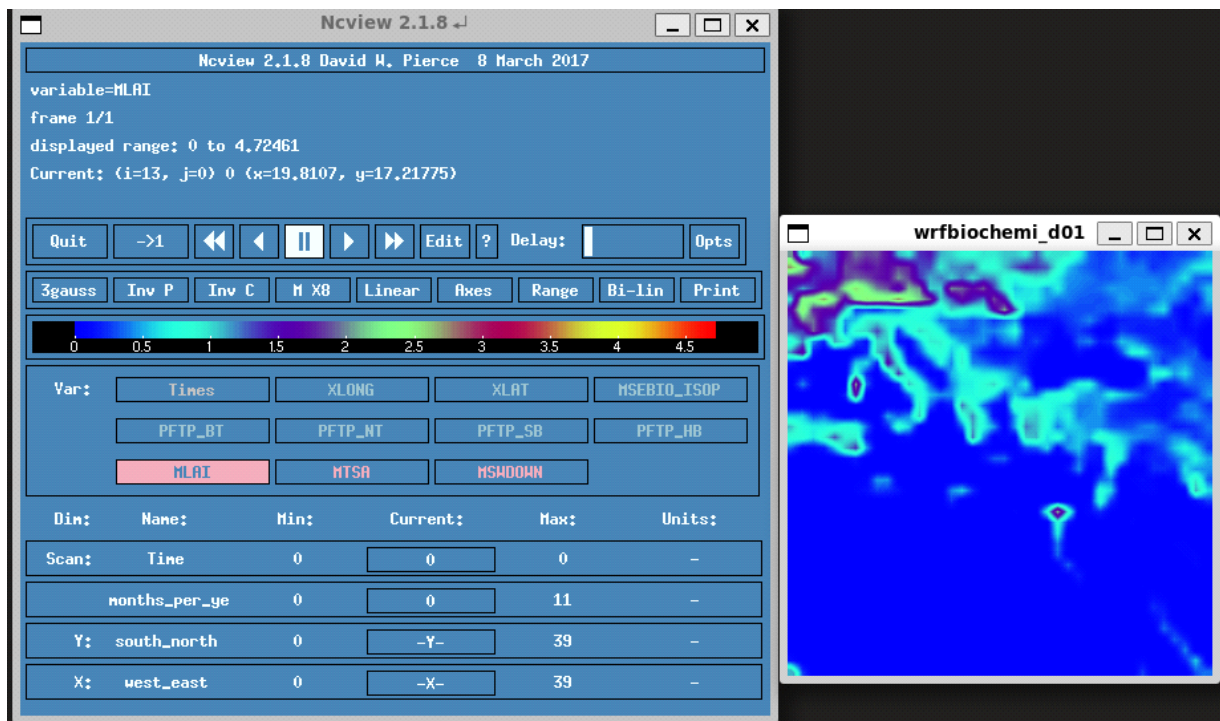
```
cd /mnt/d/Project1/code/WRF/test/em_real
ln -sf ../../../../WPS/met_em_202* ./
./real.exe #(output: wrfbdy_d01, wrfininput_d01)
```

```
-----
Domain 1: Current date being processed: 2021-01-01_00:00:00.0000, which is loop #1465 out of 1465
configflags%julyr, %julday, %gmt:      2021      1  0.00000000
d01 2021-01-01_00:00:00 Yes, this special data is acceptable to use: OUTPUT FROM METGRID V4.5
d01 2021-01-01_00:00:00 Input data is acceptable to use: met_em.d01.2021-01-01_00:00:00.nc
metgrid input_wrf.F first_date_input = 2021-01-01_00:00:00
metgrid input_wrf.F first_date_nml = 2020-01-01_00:00:00
d01 2021-01-01_00:00:00 Timing for input      0 s.
d01 2021-01-01_00:00:00 Turning off use of MAX WIND level data in vertical interpolation
d01 2021-01-01_00:00:00 Turning off use of TROPOPAUSE level data in vertical interpolation
Using sfcprsr2 to compute psfc
d01 2021-01-01_00:00:00 No average surface temperature for use with inland lakes
d01 2021-01-01_00:00:00 No soil temperature data for grid%tmn near 30 cm
d01 2021-01-01_00:00:00 Using 1 degree static annual mean temps
d01 2021-01-01_00:00:00 forcing artificial silty clay loam at      6 points, out of      1600
LAND CHANGE =      0
WATER CHANGE =      0
d01 2021-01-01_00:00:00 Timing for processing      0 s.
LBC valid between these times 2020-12-31_18:00:00 and 2021-01-01_00:00:00
d01 2021-01-01_00:00:00 Timing for output      0 s.
d01 2021-01-01_00:00:00 Timing for loop # 1465 =      0 s.
d01 2021-01-01_00:00:00 real_em: SUCCESS COMPLETE REAL_EM INIT
```

8. Deal with biogenic data:

```
cd /mnt/d/Project1/MEGAN/megan_code
./megan_bio_emiss < megan_bio_emiss.inp >& run.out #(output: wrfbiochemi_d01)

ln -sf ../../../../MEGAN/megan_code/wrfbiochemi_d01 ./
ncview wrfbiochemi_d01 #(visualization output)
```



9. Deal with chemistry data # (file namelist_chemi.input)

```
./prep_chem_sources_RADM_WRF_FIM_SIMPLE.exe #(only for one day)
```

```
=> -----
=> Gocart Background Section -----
writing spc = DMS          41          41          12
DMS for month =           1
gocart_bg_spc_name DMS
writing spc = EROD          41          41           3
writing spc = H2O2          41          41          55
writing spc = OH            41          41          55
writing spc = NO3           41          41          55
```

```
./script_prep_chem_RADM_WRF_FIM_SIMPLE.sh #(results of multiple days)
```

```
ln -sf ../../PREP-CHEM-SRC-1.5/bin/WRF-T-2020-01-01-000000-g1-bb.bin emissfire_d01
ln -sf ../../PREP-CHEM-SRC-1.5/bin/WRF-T-2020-01-01-000000-g1-ab.bin emissopt3_d01
ln -sf ../../PREP-CHEM-SRC-1.5/bin/WRF-T-2020-01-01-000000-g1-gocartBG.bin wrf_gocart_backg
ln -sf ../../PREP-CHEM-SRC-1.5/bin/WRF-T-2020-01-01-000000-g1-volc.bin volc_d01
```

```
./convert_emiss.exe #(output: wrfchemi_d01,wrffirechemi_d01,wrfchemi_gocart_bg_d01,wrfchemv_d01)
```

```
===== GO-CART BACKGROUND DATA =====
GO-CART BACKGROUND DATA file name: wrfchemi_gocart_bg_d01
I am reading global volcanic emissions, dims: = 1 40 1 40
OPENED FILE: volc_d01
now do ash mass
now do ash height
now do erup dt
OPEN EMISSIONS WRF file for emissions coming from volcano data set
I am reading global anthropogenic emissions, dims: = 1 40 1 40
OPENED FILE: emissopt3_d01
OPEN EMISSIONS WRF files for emissions coming from global data set
TRY TO OPEN FILE: emissfire_d01
calling fire emissions
Number of fire emissions: 37
FIRE EMISSIONS INPUT FILE TIME PERIOD (GMT): -1 TO 0
NOW OPEN FILE FOR WRITE: wrffirechemi_d01
FIRE EMISSIONS OUTPUT file name: wrffirechemi_d01
EMISSIONS CONVERSION : end of program
```

```
Aug 16 12:18 wrfchemi_d01*
Aug 16 12:18 wrfchemi_gocart_bg_d01*
Aug 16 12:18 wrfchemv_d01*
Aug 15 10:21 wrfem_00to12z_d01 -> ../../WRF_example_3/wrfem_00to12z*
Aug 16 12:18 wrffirechemi_d01*
```

10. Deal with boundary condition data:

Set namelist.input

Reference link: https://ruc.noaa.gov/wrf/wrf-chem/Users_guide.pdf

BEFORE BC, run real.exe to generate wrfinput_* and wrdbdy_*

Deal BC

```
cd /mnt/d/Project1/code/mozbc
./make_mozbc (export NETCDF DIR=/mnt/d/Project1/code/ioapi_env/netcdf export FC=ifort)
```

```
+++++
mozbc build Ok
+++++
```

Download dataset

```
vim MOZCART.inp #(change config file: add def_missing_var)
```

```
./mozbc < MOZCART.inp > mozbc.out #(need to change ntime)
```

```
main: setting boundary condition for num_a01
main: setting boundary condition for num_a02
main: setting boundary condition for num_a03
main: setting boundary condition for num_a04
main: setting boundary condition for oc_a01
main: setting boundary condition for oc_a02
successfully exited from module_wrfchem_lib ...
successfully exited from module_mozart_lib ...
bc_wrfchem completed successfull
```

11. Deal with anthropogenic data

```
cd /mnt/d/Project1/code/ANTHRO/src
./make_anthro #(export FC=ifort)
```

```
+++++
anthro_emis build Ok
+++++
```

```
./anthro_emis < anthro_emis.inp > anthro_emis.out #(NOTICE: *.inp & *.nc file)
```

```
get_src_time_ndx; src_dir,src_fn =
/mnt/d/Project1/data/Volcano/Global_emissions_v3/EMISSION_DATA/EDGARV7/v7.0_FT2
021_CH4_2020_2021_TOTALS.0.1x0.1.nc
get_src_time_ndx; interp_date,datesec,ntimes =      20200101      0
2
get_src_time_ndx; tndx =      1
map_src_emissions: count of points <,> data min,max lat =      0
xndx_src =      1208      1221
map_src_emissions: count of points < data min lat =      0
map_src_emissions: count of points > data max lat =      0
yndx_src =      550      561

-----
anthro_emis completed successfully
-----
```

Output file

```
wrfchemi_00z_d01*
wrfchemi_12z_d01*
```

12. Generate WRF output (NEED TO CHANGE VERSION4)

```
ln -sf ../../../../ANTHRO/src/wrfchemi_00z_d01 wrfchemi_d01_2020-01-01_00:00:00
./real.exe
```

```
d01 2021-01-01_00:00:00 Timing for processing      0 s.
LBC valid between these times 2020-12-31_18:00:00.0000 2021-01-01_00:00:00
d01 2021-01-01_00:00:00 Timing for output      0 s.
d01 2021-01-01_00:00:00 Timing for loop # 1465 =      0 s.
d01 2021-01-01_00:00:00 real_em: SUCCESS COMPLETE REAL_EM INIT
```

```
./wrf.exe
```

```

Timing for main: time 2020-01-01_00:00:00 on domain 1: 0.63958 elapsed seconds
Timing for main: time 2020-01-01_00:57:00 on domain 1: 0.64313 elapsed seconds
Timing for main: time 2020-01-01_00:58:00 on domain 1: 0.63745 elapsed seconds
Timing for main: time 2020-01-01_00:59:00 on domain 1: 0.64123 elapsed seconds
60 30
Timing for main: time 2020-01-01_01:00:00 on domain 1: 0.66976 elapsed seconds
Timing for Writing wrfout_d01_2020-01-01_01:00:00 for domain 1: 0.97401 elapsed seconds
Timing for main: time 2020-01-01_01:01:00 on domain 1: 3.43285 elapsed seconds
Timing for main: time 2020-01-01_01:02:00 on domain 1: 0.64792 elapsed seconds
Timing for main: time 2020-01-01_01:03:00 on domain 1: 0.68209 elapsed seconds
Timing for main: time 2020-01-01_01:04:00 on domain 1: 0.70582 elapsed seconds

```